

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0611

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions
– Case Based Learning **(BL 5)**

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks:50

Annexure A

Comparative Analysis

Code of Conduct:

*I P.Manideep(192212252)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Short Answer Test

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.
3. Analyze the differences between K-Nearest Neighbour and Radial Basis Function models in classification problems. Provide an example and compare decision boundaries, training time, and generalization ability.
4. Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.
6. Compare Naïve Bayes and Logistic Regression for a text classification task such as spam detection. Analyze assumptions about feature independence, probability estimation, computational efficiency, and performance on high-dimensional data. Discuss scenarios where one method outperforms the other.
10. Compare Principal Component Analysis (PCA) and Feature Selection techniques for dimensionality reduction. Analyze differences in information preservation, computational complexity, interpretability, and impact on model performance. Discuss how each approach affects downstream machine learning tasks.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	15	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	20	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

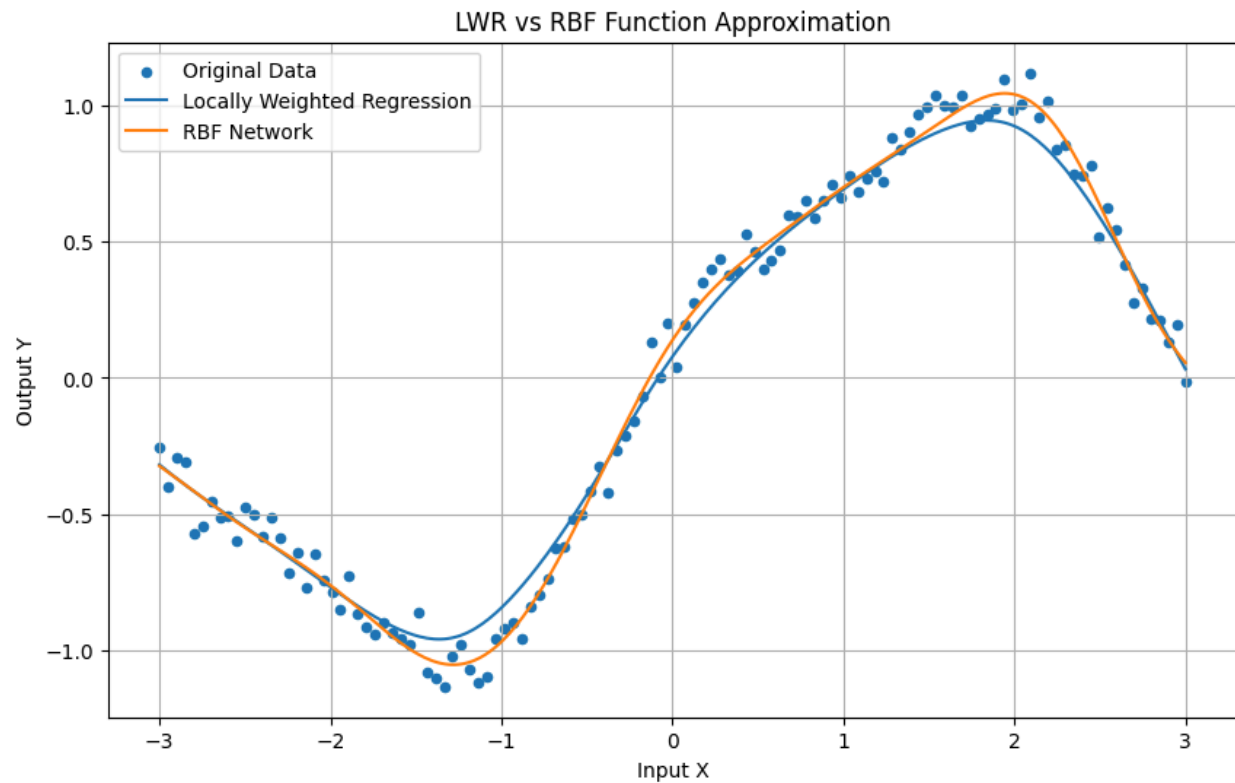
Q2

OUTPUT:

===== PERFORMANCE COMPARISON =====

LWR Mean Squared Error : 0.008302919494733538

RBF Mean Squared Error : 0.0043239147440450095



Q3

OUTPUT:

===== ACCURACY =====

KNN Accuracy : 0.861111111111112

RBF Accuracy : 0.902777777777778

===== KNN REPORT =====

	precision	recall	f1-score	support
0	0.78	0.84	0.81	25
1	0.91	0.87	0.89	47
accuracy			0.86	72
macro avg	0.84	0.86	0.85	72
weighted avg	0.86	0.86	0.86	72

```

===== RBF REPORT =====
precision recall f1-score support

```

```

0    0.82    0.92    0.87    25
1    0.95    0.89    0.92    47

```

```

accuracy          0.90    72
macro avg         0.89    0.91    0.90    72
weighted avg      0.91    0.90    0.90    72

```

KNN Confusion Matrix:

```

[[21  4]
 [ 6 41]]

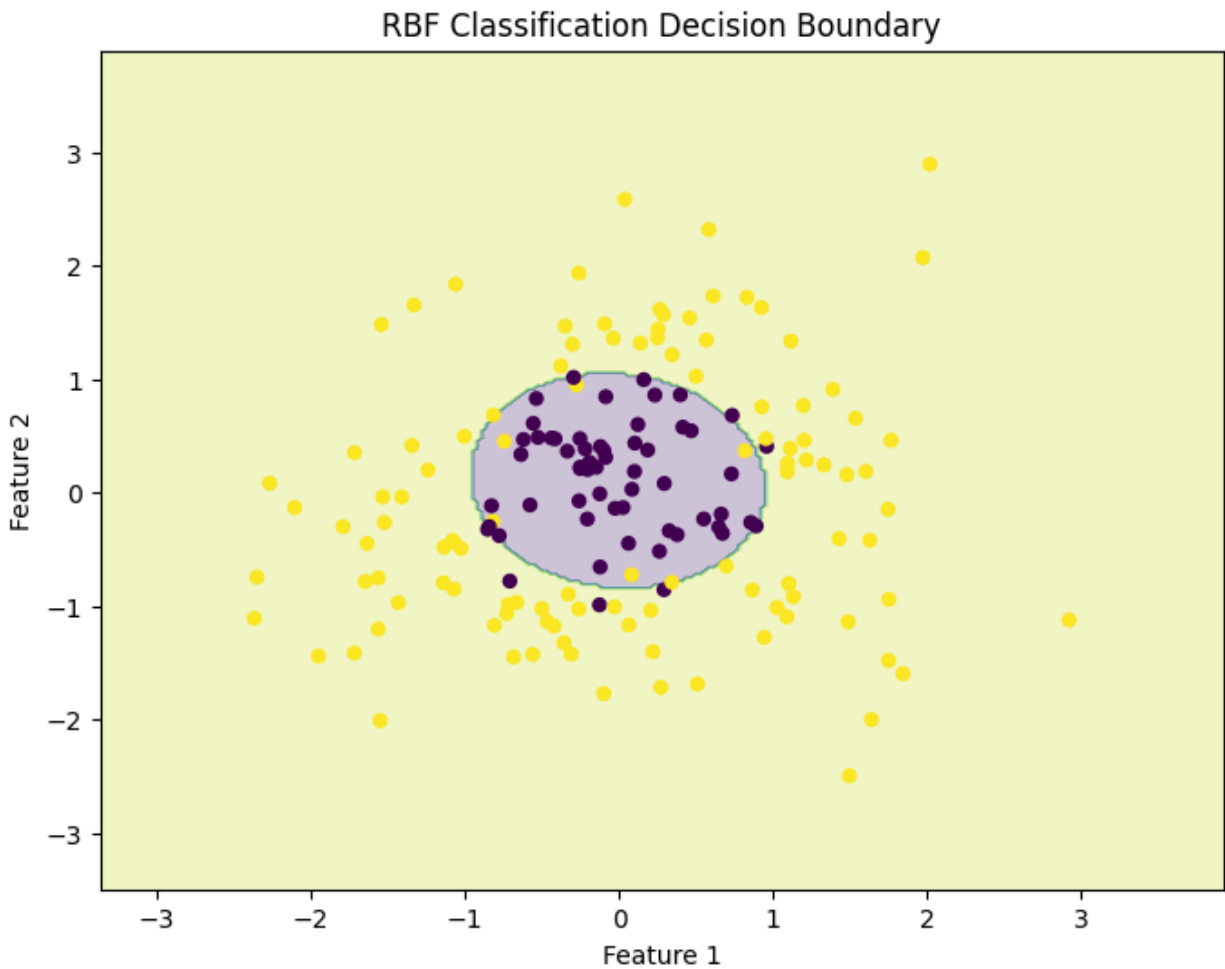
```

RBF Confusion Matrix:

```

[[23  2]
 [ 5 42]]

```



Q4

OUTPUT:

===== CASE-BASED LEARNING =====

Similar User IDs: [18, 28, 16, 21, 17]

Previous Ratings: [5.0, 5.0, 5.0, 5.0, 5.0]

Predicted Rating: 5.0

===== LOCALLY WEIGHTED REGRESSION =====

Predicted Rating: 4.99

===== COMPARISON =====

Case-Based Learning : 5.0

Locally Weighted Regression : 4.99

Case-Based Learning retrieves similar historical cases.

LWR assigns continuous weights based on similarity.

Q6

OUTPUT:

===== ACCURACY =====

Naive Bayes Accuracy: 1.0

Logistic Regression Accuracy: 1.0

===== NAIVE BAYES =====

	precision	recall	f1-score	support
ham	1.00	1.00	1.00	20
spam	1.00	1.00	1.00	20
accuracy			1.00	40
macro avg	1.00	1.00	1.00	40
weighted avg	1.00	1.00	1.00	40

===== LOGISTIC REGRESSION =====

	precision	recall	f1-score	support
ham	1.00	1.00	1.00	20
spam	1.00	1.00	1.00	20
accuracy			1.00	40

macro avg	1.00	1.00	1.00	40
weighted avg	1.00	1.00	1.00	40

Naive Bayes Confusion Matrix:

```
[[20  0]
 [ 0 20]]
```

Logistic Regression Confusion Matrix:

```
[[20  0]
 [ 0 20]]
```

===== NEW MESSAGE PREDICTIONS =====

Message: Congratulations you won a free prize

Naive Bayes: spam

Logistic Regression: spam

Message: Please attend the meeting tomorrow

Naive Bayes: ham

Logistic Regression: ham

Message: Claim your free money now

Naive Bayes: spam

Logistic Regression: spam

Message: Please submit the assignment

Naive Bayes: ham

Logistic Regression: ham

Q10

OUTPUT:

===== PERFORMANCE =====

PCA R2 Score: 0.7213

PCA RMSE: 0.8804

Feature Selection R2 Score: 0.8045

Feature Selection RMSE: 0.7373

Selected Features:

```
['feature1', 'feature3']
```

PCA Explained Variance Ratio:

```
[0.40835863 0.30933726]
```

Total Variance Preserved: 0.7177

